

Albert (Geyang) Xu

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EDUCATION

University of California, Riverside

Ph.D. in Computer Science (ML Systems Reliability · Advisor: Prof. Qian Zhang)

Sep 2026 – Present

Riverside, CA

University of California, San Diego

M.S. in Computer Science (GPA: 3.88/4.0)

Sep 2022 – Mar 2024

La Jolla, CA

University of Liverpool

B.S. in Computer Science (GPA: 3.93/4.0, top 3%)

Sep 2018 – Jun 2022

Liverpool, UK

PUBLICATION

J. Zhu, G. Xu, et al. *Stress-Testing ML Pipelines with Adversarial Data Corruption*. PVLDB 18(11), 2025. [\[DOI\]](#) · [\[Code\]](#)

EXPERIENCE

Fuyao Glass America Inc.

Software Engineer

Oct 2025 – Apr 2026

Moraine, OH

- Fine-tuned **YOLOv8** for inline automotive-glass inspection on labeled line-camera frames, reaching **mAP@0.5** $\approx$ **0.87** and **>90%** recall on held-out images under plant miss-rate targets.
- Productionized defect detection on line-camera frames with cron, alerts, and access control — **99.3%** job success and **2.4 hrs/week** less downtime; ran A/B and DOE setpoint studies in Jupyter for process tuning.
- Built **Python/SQL** pipelines that fuse defect detections with coating, tempering, and inspection signals into a shared analytics layer; shipped role-based dashboards tracking OEE, FPY, scrap, and changeover to support high-efficiency production.

4Pexonic Inc.

Software Engineer

Oct 2024 – Oct 2025

San Diego, CA

- Developed an academic impact analysis platform **mining citation patterns** across tens of thousands of records using Node.js, MongoDB, and Next.js with Docker, serving university researchers and policy institutions.
- Built two-tier caching (in-process **LRU** + **Redis**), cutting median and p95 latency by **40%**.
- Integrated **gRPC** microservices; implemented bcrypt/session auth and Nginx TLS.

UCSD Halicioğlu Data Science Institute

Research Engineer

Oct 2023 – Oct 2024

La Jolla, CA

- Built an Injector pipeline (generate → sample → inject → eval) driving a gradient-free **bi-level search**: **beam search** over corruption dependency graphs at the outer level, **Optuna/TPE** over parameters at the inner, with the pipeline as a black box.
- Benchmarked the search against random corruption injection, cutting downstream model **AUC** by **>0.25** and turning each win into a readable failure pattern — which subpopulation broke, and by what mechanism.
- Ran **Inject** → **Clean** → **Retrain** benchmarks across missing-value, selection-bias, and outlier scenarios to evaluate state-of-the-art cleaning, debiasing, and UQ pipelines and expose their data-centric robustness gaps.

PROJECTS

SurfStore: Distributed Cloud Storage Server Project (Go, gRPC, RAFT) | [Link](#)

- Designed and implemented a Dropbox-like distributed file storage service in **Go**, using consistent hashing to place blocks across multiple **BlockStores** and **gRPC** for client-metadata communication.
- Enhanced fault tolerance by implementing the **RAFT** consensus protocol over the metadata service, keeping the store available and consistent through simulated leader crashes and network partitions.

TECHNICAL SKILLS

Languages: Python, Java, Go, SQL, C/C++, JavaScript, R

Databases: PostgreSQL (pgvector), MongoDB, Redis

Frameworks and Tools: Docker, Git, gRPC, Node.js, Next.js, Django, React, Optuna, Jupyter, YOLOv8, AWS, LangGraph